

STAT 153 & 248 - Time Series

Lecture Eight

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1 Recap from last lecture

In the last lecture, we considered fitting the sinusoidal model

$$y_t = \beta_0 + \beta_1 \cos(2\pi ft) + \beta_2 \sin(2\pi ft) + \epsilon_t \quad \text{with } \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \quad (1)$$

to observed time series $y_1, \dots, y_n$. A key role for inferring the frequency parameter f in this model is played by:

$$RSS(f) := \min_{\beta_0, \beta_1, \beta_2} \sum_{t=1}^n (y_t - \beta_0 - \beta_1 \cos(2\pi ft) - \beta_2 \sin(2\pi ft))^2 = \|y - X_f \hat{\beta}_f\|^2$$

where

$$y = \begin{pmatrix} y_1 \\ \cdot \\ \cdot \\ \cdot \\ y_n \end{pmatrix} \quad \text{and} \quad X_f = \begin{pmatrix} 1 & \cos(2\pi f(1)) & \sin(2\pi f(1)) \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ 1 & \cos(2\pi f(n)) & \sin(2\pi f(n)) \end{pmatrix} \quad \text{and} \quad \hat{\beta}_f = (X_f^T X_f)^{-1} X_f^T y.$$

$RSS(f)$ is simply the residual sum of squares in the linear regression model obtained by fixing the frequency parameter f . It describes how well the sinusoid with frequency f fits the observed data $y_1, \dots, y_n$.

Calculating $RSS(f)$ separately for each value of f in a grid of values of f can be computationally quite inefficient when n is large (for example, in the audio dataset, n is of the order of 10^5 ; in this case for calculating $RSS(f)$ separately for f in a grid of length 10^4 seems to take at least 4 full minutes). In practice, in such situations, one uses an efficient method for calculating $RSS(f)$ based on the Fast Fourier Transform (FFT) algorithm. We shall see the main ideas behind this method today (without going into the details of FFT).

2 Fourier Frequencies

For computing $RSS(f)$, we need a grid of values for f . The most commonly used grid is given by:

$$\mathcal{F} := \left\{ 0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{[n/2]}{n} \right\} \quad (2)$$

where $\lfloor n/2 \rfloor$ is the largest integer smaller than or equal to $n/2$.

A frequency of the form j/n where $j \in \{0, 1, 2, \dots, n-1\}$ and n is the observed data size is called a **Fourier Frequency**. So the grid (2) consists of all Fourier frequencies that are in the range $[0, 1/2]$.

The main reason for taking the grid to consist of Fourier Frequencies is that $RSS(f)$, $f \in \mathcal{F}$ can be computed very efficiently (in time $O(n \log n)$) using a classical algorithm known as the Fast Fourier Transform (FFT). We explain the high level details behind this fact today (without going into the workings of the FFT algorithm).

3 Formula for $RSS(f)$ when f is a Fourier Frequency

When f is a Fourier Frequency, one can write down a more explicit formula for $RSS(f)$. For this, first note that for every f :

$$RSS(f) = \min_{\beta} \|y - X_f \beta\|^2 = \|y - X_f \hat{\beta}_f\|^2 \quad \text{where } \hat{\beta}_f := (X_f^T X_f)^{-1} X_f^T y.$$

Thus

$$\begin{aligned} RSS(f) &= \|y - X_f \hat{\beta}_f\|^2 \\ &= (y - X_f \hat{\beta}_f)^T (y - X_f \hat{\beta}_f) = y^T y - \hat{\beta}_f^T X_f^T y - y^T X_f \hat{\beta}_f + \hat{\beta}_f^T X_f^T X_f \hat{\beta}_f. \end{aligned}$$

Plugging in $\hat{\beta}_f = (X_f^T X_f)^{-1} X_f^T y$ above, we get

$$\begin{aligned} RSS(f) &= y^T y - y^T X_f (X_f^T X_f)^{-1} X_f^T y - y^T X_f (X_f^T X_f)^{-1} X_f^T y \\ &\quad + y^T X_f (X_f^T X_f)^{-1} X_f^T X_f (X_f^T X_f)^{-1} X_f^T y \\ &= y^T y - y^T X_f (X_f^T X_f)^{-1} X_f^T y - y^T X_f (X_f^T X_f)^{-1} X_f^T y \\ &\quad + y^T X_f (X_f^T X_f)^{-1} X_f^T y \\ &= y^T y - y^T X_f (X_f^T X_f)^{-1} X_f^T y \end{aligned}$$

Now

$$X_f^T X_f = \begin{pmatrix} n & \sum_{t=1}^n \cos(2\pi ft) & \sum_{t=1}^n \sin(2\pi ft) \\ \sum_{t=1}^n \cos(2\pi ft) & \sum_{t=1}^n \cos^2(2\pi ft) & \sum_{t=1}^n \cos(2\pi ft) \sin(2\pi ft) \\ \sum_{t=1}^n \sin(2\pi ft) & \sum_{t=1}^n \cos(2\pi ft) \sin(2\pi ft) & \sum_{t=1}^n \sin^2(2\pi ft) \end{pmatrix}$$

Now suppose that $f \in (0, 0.5)$ and suppose that f is a Fourier frequency i.e., it is of the form $f = j/n$ for some integer j . Then it turns out that

$$\begin{aligned} \sum_{t=1}^n \cos(2\pi ft) &= 0 & \sum_{t=1}^n \sin(2\pi ft) &= 0 \\ \sum_{t=1}^n \cos^2(2\pi ft) &= \frac{n}{2} & \sum_{t=1}^n \sin^2(2\pi ft) &= \frac{n}{2} \\ \sum_{t=1}^n \cos(2\pi ft) \sin(2\pi ft) &= 0 \end{aligned} \tag{3}$$

These facts are intuitively obvious. For a formal proof, see Subsection 3.1.

As a result, for such f ,

$$X_f^T X_f = \begin{pmatrix} n & 0 & 0 \\ 0 & n/2 & 0 \\ 0 & 0 & n/2 \end{pmatrix} \quad \text{so that} \quad (X_f^T X_f)^{-1} = \begin{pmatrix} 1/n & 0 & 0 \\ 0 & 2/n & 0 \\ 0 & 0 & 2/n \end{pmatrix}$$

This gives

$$\begin{aligned} & y^T X_f (X_f^T X_f)^{-1} X_f^T y \\ &= y^T \begin{pmatrix} 1 & \cos(2\pi f(1)) & \sin(2\pi f(1)) \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ 1 & \cos(2\pi f(n)) & \sin(2\pi f(n)) \end{pmatrix} \begin{pmatrix} 1/n & 0 & 0 \\ 0 & 2/n & 0 \\ 0 & 0 & 2/n \end{pmatrix} \begin{pmatrix} 1 & \cdot & \cdot & \cdot & 1 \\ \cos(2\pi f(1)) & \cdot & \cdot & \cdot & \cos(2\pi f(n)) \\ \sin(2\pi f(1)) & \cdot & \cdot & \cdot & \sin(2\pi f(n)) \end{pmatrix} y \\ &= (\sum_t y_t \quad \sum_t y_t \cos(2\pi ft) \quad \sum_t y_t \sin(2\pi ft)) \begin{pmatrix} 1/n & 0 & 0 \\ 0 & 2/n & 0 \\ 0 & 0 & 2/n \end{pmatrix} \begin{pmatrix} \sum_t y_t \\ \sum_t y_t \cos(2\pi ft) \\ \sum_t y_t \sin(2\pi ft) \end{pmatrix} \\ &= \frac{1}{n} \left(\sum_t y_t \right)^2 + \frac{2}{n} \left(\sum_t y_t \cos(2\pi ft) \right)^2 + \frac{2}{n} \left(\sum_t y_t \sin(2\pi ft) \right)^2 \\ &= n\bar{y}^2 + \frac{2}{n} \left(\sum_t y_t \cos(2\pi ft) \right)^2 + \frac{2}{n} \left(\sum_t y_t \sin(2\pi ft) \right)^2. \end{aligned}$$

Therefore for Fourier frequencies in the range $(0, 0.5)$, we get

$$RSS(f) = y^T y - n\bar{y}^2 - \frac{2}{n} \left(\sum_t y_t \cos(2\pi ft) \right)^2 - \frac{2}{n} \left(\sum_t y_t \sin(2\pi ft) \right)^2$$

or equivalently

$$RSS(f) = \sum_t (y_t - \bar{y})^2 - \frac{2}{n} \left(\sum_t y_t \cos(2\pi ft) \right)^2 - \frac{2}{n} \left(\sum_t y_t \sin(2\pi ft) \right)^2 \quad (4)$$

When $f = 0$ and $f = 1/2$, the above formula needs to be slightly modified. When $f = 0$, the sinusoidal model (1) simply becomes:

$$y_t = \beta_0 + \beta_1 + \epsilon_t \quad \text{with } \epsilon_t \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2).$$

There is only one effective parameter coefficient parameter $(\beta_0 + \beta_1)$ here which will be estimated by $\bar{y}$ so that RSS becomes

$$RSS(0) = \sum_t (y_t - \bar{y})^2. \quad (5)$$

When $f = 1/2$ and n is even (when n is odd, $1/2$ cannot be a Fourier frequency so it will not be considered), the model (1) becomes

$$y_t = \beta_0 + \beta_1 \cos(\pi t) + \epsilon_t = \beta_0 + \beta_1 (-1)^t + \epsilon_t.$$

For this model, it is easy to check that

$$X^T X = \begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}$$

so that

$$RSS(1/2) = \sum_t (y_t - \bar{y})^2 - \frac{1}{n} \left(\sum_t y_t (-1)^t \right)^2 \quad (6)$$

Let us not worry too much about the edge cases $f = 0$ and $f = 1/2$, and focus on the formula (4). Note again that this formula holds whenever $f \in (0, 0.5)$ and f is a Fourier frequency (i.e., nf is an integer).

3.1 Proof of the identities in (3)

The identities (3) (which hold when f is a Fourier frequency) are all intuitively obvious. For completeness, a formal proof is given below.

Note that $0 < f < 1/2$ and that nf is an integer.

$$\begin{aligned} & \sum_{t=1}^n \cos(2\pi ft) \\ &= \frac{1}{2} \sum_{t=1}^n e^{2\pi i ft} + \frac{1}{2} \sum_{t=1}^n e^{-2\pi i ft} \\ &= \frac{1}{2} \frac{e^{2\pi i f}}{e^{2\pi i f} - 1} (e^{2\pi i n f} - 1) + \frac{1}{2} \frac{e^{-2\pi i f}}{e^{-2\pi i f} - 1} (e^{-2\pi i n f} - 1) \\ &= \frac{1}{2} \frac{e^{2\pi i f}}{e^{2\pi i f} - 1} (\cos(2\pi n f) - 1 + i \sin(2\pi n f)) + \frac{1}{2} \frac{e^{-2\pi i f}}{e^{-2\pi i f} - 1} (\cos(2\pi n f) - 1 - i \sin(2\pi n f)) \\ &= 0 \end{aligned}$$

because $\cos(2\pi n f) = \cos(2\pi(\text{integer})) = 1$ and $\sin(2\pi n f) = \sin(2\pi(\text{integer})) = 0$.

Similarly

$$\begin{aligned} & \sum_{t=1}^n \sin(2\pi ft) \\ &= \frac{1}{2i} \sum_{t=1}^n e^{2\pi i ft} - \frac{1}{2i} \sum_{t=1}^n e^{-2\pi i ft} \\ &= \frac{1}{2i} \frac{e^{2\pi i f}}{e^{2\pi i f} - 1} (e^{2\pi i n f} - 1) - \frac{1}{2i} \frac{e^{-2\pi i f}}{e^{-2\pi i f} - 1} (e^{-2\pi i n f} - 1) \\ &= \frac{1}{2i} \frac{e^{2\pi i f}}{e^{2\pi i f} - 1} (\cos(2\pi n f) - 1 + i \sin(2\pi n f)) - \frac{1}{2i} \frac{e^{-2\pi i f}}{e^{-2\pi i f} - 1} (\cos(2\pi n f) - 1 - i \sin(2\pi n f)) \\ &= 0 \end{aligned}$$

Next note

$$\sum_{t=1}^n \cos^2(2\pi ft) = \sum_{t=1}^n \frac{1 + \cos(4\pi ft)}{2} = \frac{n}{2} + \frac{1}{2} \sum_{t=1}^n \cos(4\pi ft)$$

and

$$\sum_{t=1}^n \sin^2(2\pi ft) = \sum_{t=1}^n \frac{1 - \cos(4\pi ft)}{2} = \frac{n}{2} - \frac{1}{2} \sum_{t=1}^n \cos(4\pi ft).$$

The quantity $\sum_t \cos(4\pi ft)$ which appears in both the above terms turns out to be zero because

$$\begin{aligned}
& \sum_{t=1}^n \cos(4\pi ft) \\
&= \frac{1}{2} \sum_{t=1}^n e^{4\pi i f t} + \frac{1}{2} \sum_{t=1}^n e^{-4\pi i f t} \\
&= \frac{1}{2} \frac{e^{4\pi i f}}{e^{4\pi i f} - 1} (e^{4\pi i n f} - 1) + \frac{1}{2} \frac{e^{-4\pi i f}}{e^{-4\pi i f} - 1} (e^{-4\pi i n f} - 1) \\
&= \frac{1}{2} \frac{e^{4\pi i f}}{e^{4\pi i f} - 1} (\cos(4\pi n f) - 1 + i \sin(4\pi n f)) + \frac{1}{2} \frac{e^{-4\pi i f}}{e^{-4\pi i f} - 1} (\cos(4\pi n f) - 1 - i \sin(4\pi n f)) \\
&= 0
\end{aligned}$$

because $\cos(4\pi n f) = \cos(4\pi(\text{integer})) = 1$ and $\sin(4\pi n f) = \sin(4\pi(\text{integer})) = 0$.

Finally

$$\begin{aligned}
& \sum_{t=1}^n \cos(2\pi f t) \sin(2\pi f t) \\
&= \sum_{t=1}^n \sin(4\pi f t) \\
&= \frac{1}{2i} \sum_{t=1}^n e^{4\pi i f t} - \frac{1}{2i} \sum_{t=1}^n e^{-4\pi i f t} \\
&= \frac{1}{2i} \frac{e^{4\pi i f}}{e^{4\pi i f} - 1} (e^{4\pi i n f} - 1) - \frac{1}{2i} \frac{e^{-4\pi i f}}{e^{-4\pi i f} - 1} (e^{-4\pi i n f} - 1) \\
&= \frac{1}{2i} \frac{e^{4\pi i f}}{e^{4\pi i f} - 1} (\cos(4\pi n f) - 1 + i \sin(4\pi n f)) - \frac{1}{2i} \frac{e^{-4\pi i f}}{e^{-4\pi i f} - 1} (\cos(4\pi n f) - 1 - i \sin(4\pi n f)) \\
&= 0
\end{aligned}$$

The intermediate calculations above include the terms $e^{2\pi i f} - 1$ and $e^{4\pi i f} - 1$ in the denominators. We need to make sure that these terms are not zero (otherwise the above proofs would not be valid).

$$e^{2\pi i f} - 1 = \cos(2\pi f) - 1 + i \sin(2\pi f)$$

which cannot be zero because $\cos(2\pi f) < 1$ for $f \in (0, 0.5)$, and

$$e^{4\pi i f} - 1 = \cos(4\pi f) - 1 + i \sin(4\pi f)$$

which also cannot be zero because $\cos(4\pi f) < 1$ for $f \in (0, 0.5)$.

4 The Periodogram

Given a time series dataset $y_1, \dots, y_n$, its **periodogram** is the function $I(f)$, $0 < f < 1/2$, defined as follows:

$$I(f) := \frac{1}{n} \left(\sum_{t=1}^n y_t \cos(2\pi f t) \right)^2 + \frac{1}{n} \left(\sum_{t=1}^n y_t \sin(2\pi f t) \right)^2 \quad \text{for } f \in (0, 0.5)$$

From the formula (4), we have

$$RSS(f) = \sum_t (y_t - \bar{y})^2 - 2I(f) \quad \text{when } f \in (0, 0.5) \text{ is a Fourier Frequency.}$$

The periodogram is closely related to the Discrete Fourier Transform (DFT) of the data y_t . This is explained next.

5 Discrete Fourier Transform (DFT) and Periodogram

The periodogram $I(f)$ clearly can be rewritten as:

$$I(f) = \frac{1}{n} \left| \sum_{t=1}^n y_t \exp(-2\pi i f t) \right|^2.$$

The term inside the modulus sign above is almost the same as the Discrete Fourier Transform (DFT) of the data. There is only one difference which comes from changing the data indexing slightly. While discussing the DFT, it is a standard convention to write the data as $y_0, y_1, \dots, y_{n-1}$ (instead of $y_1, \dots, y_n$). In other words, we start the indexing with 0 (as in Python) as opposed to 1. With this indexing, the DFT is defined as follows.

The DFT of data $y_0, \dots, y_{n-1}$ is defined by:

$$b_j := \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i j t}{n}\right) \quad \text{for } j = 0, 1, \dots, n-1. \quad (7)$$

In words, b_j is the dot product between the data and the complex sinusoid with frequency $f = j/n$. Note that the complex sinusoid with frequency f is given by:

$$\exp(2\pi i f t) \quad \text{for } t = 0, 1, 2, \dots, n-1.$$

The n (possibly) complex numbers $b_0, b_1, \dots, b_{n-1}$ are collectively called the DFT of $y_0, \dots, y_{n-1}$. Typically, $y_0, \dots, y_{n-1}$ will represent observed time series data. It is important to note that even though $y_0, \dots, y_{n-1}$ are real-valued, their DFT $b_0, \dots, b_{n-1}$ can be complex-valued.

More on indexing: As mentioned previously, while defining the DFT, we index the data starting from zero. In the definition (7), the first data point (which is y_0) is being multiplied by $\exp(-2\pi i j(0)/n) = 1$, the second data point is being multiplied by $\exp(-2\pi i j/n)$ and, in general, the t -th data point is being multiplied by $\exp(-2\pi i j(t-1)/n)$. If we instead define the DFT by:

$$\tilde{b}_j := \sum_{t=1}^n y_t \exp\left(-\frac{2\pi i j t}{n}\right),$$

then the t -th data point will be multiplied by $\exp(-2\pi i j t/n)$. b_j and $\tilde{b}_j$ will be different and they will be related by:

$$b_j = \tilde{b}_j \exp\left(\frac{2\pi i j}{n}\right).$$

The multiplier $\exp(2\pi i j/n)$ above has modulus one which means that

$$|b_j| = |\tilde{b}_j|.$$

So if we are only looking at the moduli of the DFT terms (note the periodogram only involves the moduli of DFT), then it does not matter whether we index data starting with 0 or 1. The standard convention while studying the DFT is to index starting from 0.

The connection between the periodogram and the DFT is given by:

$$I(j/n) = \frac{|b_j|^2}{n} \quad \text{for } 0 < \frac{j}{n} < 1/2$$

and the connection between $RSS(j/n)$ and DFT is given by:

$$RSS(j/n) = \sum_t (y_t - \bar{y})^2 - 2I(j/n) \quad \text{for } 0 > j/n > 0.5.$$

It is important to note that the DFT can be calculated very efficiently. The DFT of $y = (y_0, \dots, y_{n-1})^T$ can be obtained in `numpy` using the command `np.fft.fft(y)`. Here `fft` stands for Fast Fourier Transform which is a special efficient algorithm for computing the DFT. Naively, it would seem that to compute the DFT would need $O(n^2)$ computation (for each of n values of j , we have to compute b_j which requires a sum over the dataset of size n), but the FFT algorithm exploits symmetry and redundancy in the complex exponentials to compute the DFT in only $O(n \log n)$ time. We will not be going over the details of the FFT algorithm in class.